

FINANCIAL TIME SERIES ANALYSIS

VICTOR XIA - STATISTICS VXIA002@UCR.EDU

JOHN PLEINES - MENTOR, DEPARTMENT OF STATISTICS JPLEI003@UCR.EDU

Background

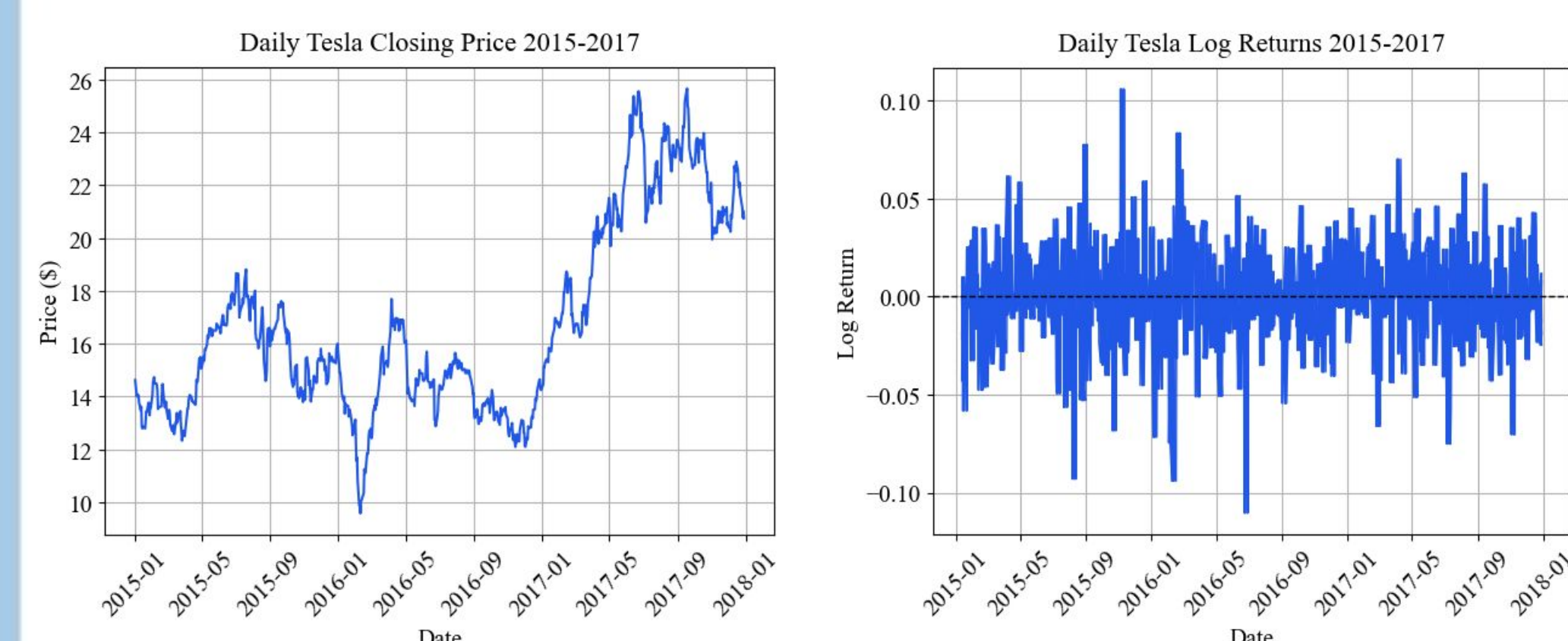
Objective:

- To evaluate common time series models and their trading performance on real financial data.
- To explore real-world financial data behavior.



Stock ticker data is a type of time series data commonly used in real life applications in the financial sector and day trading, it is also notoriously difficult to forecast due to market complexity and how dynamically impacted the data is by a multitude of variables. Our SMART Project centers around analyzing financial time series data by investigating well-known stock tickers and their unique data behaviors.

Stationarity



Many time series models require that the data is stationary, meaning that the data has a constant mean and variance. The Tesla closing prices above exhibit non-stationarity due their shifting mean and variance over time. In order to remedy this issue, the data sets were transformed to log returns (right plot above) by taking the log of the ratio between the present day's closing price and previous day's closing price. The Augmented Dickey Fuller (ADF) test was used to help verify that our transformed log return data exhibits stationarity.

Models

- The Autoregressive (AR) model is one of the simplest time series models, using the first p past values as regressors to model the next value.
- The AR(p) model is defined below:

$$X_t = \sum_{i=1}^p \beta_i X_{t-i} + \epsilon_t$$

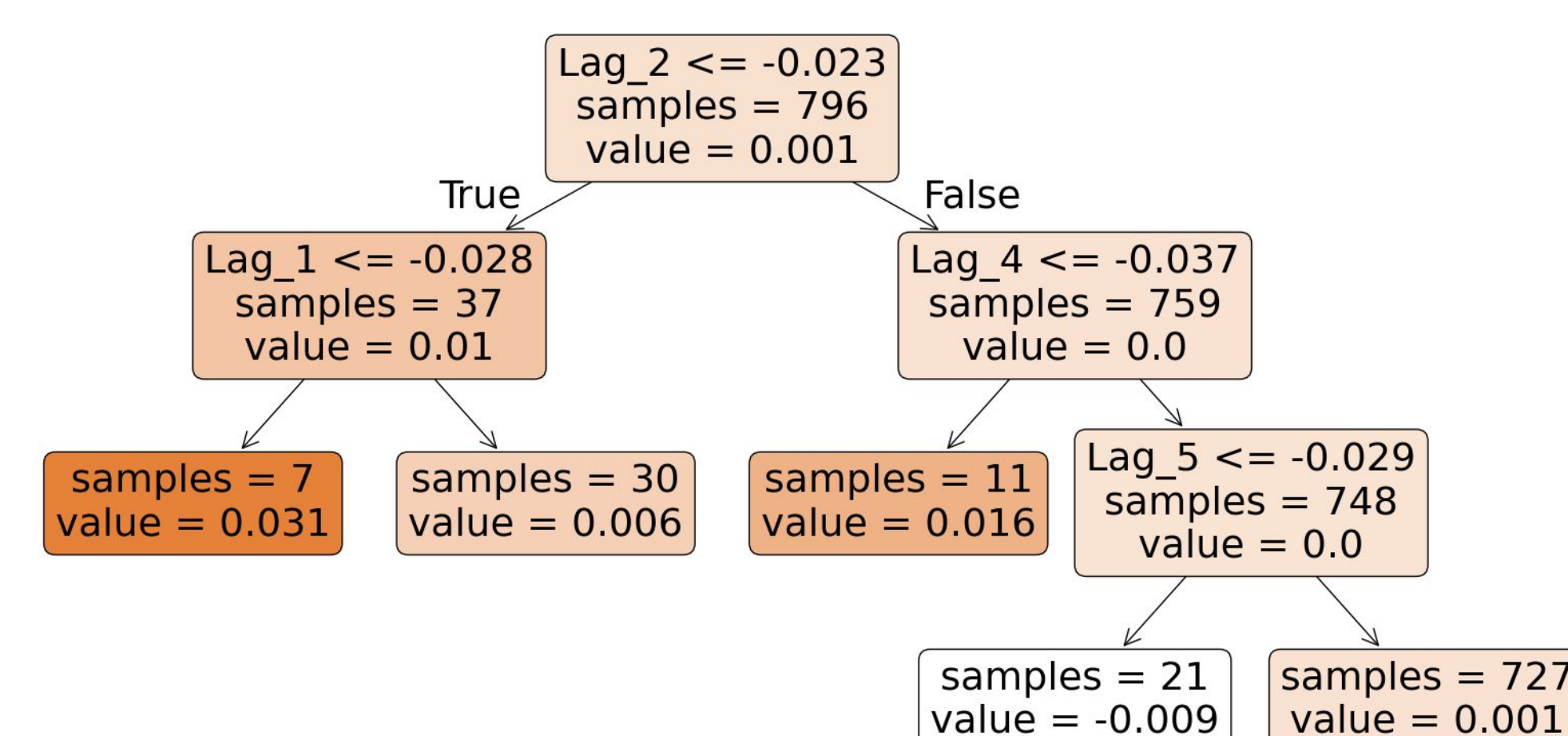
- The Autoregressive Moving Average (ARMA) model builds off of the AR(p) model by additionally incorporating the first past q error terms as regressors as the moving average component. This makes it more robust and flexible for modeling short term shocks.
- The ARMA(p, q) model is defined below:

$$X_t = \sum_{i=1}^p \beta_i X_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t$$

- The Seasonal Autoregressive Moving Average (SARMA) model extends the ARMA model by incorporating seasonality via P seasonal AR lags and Q seasonal MA lags, where s is the number of time steps that make up a seasonal cycle.
- The SARMA (p, q, P, Q, s) model is defined below:

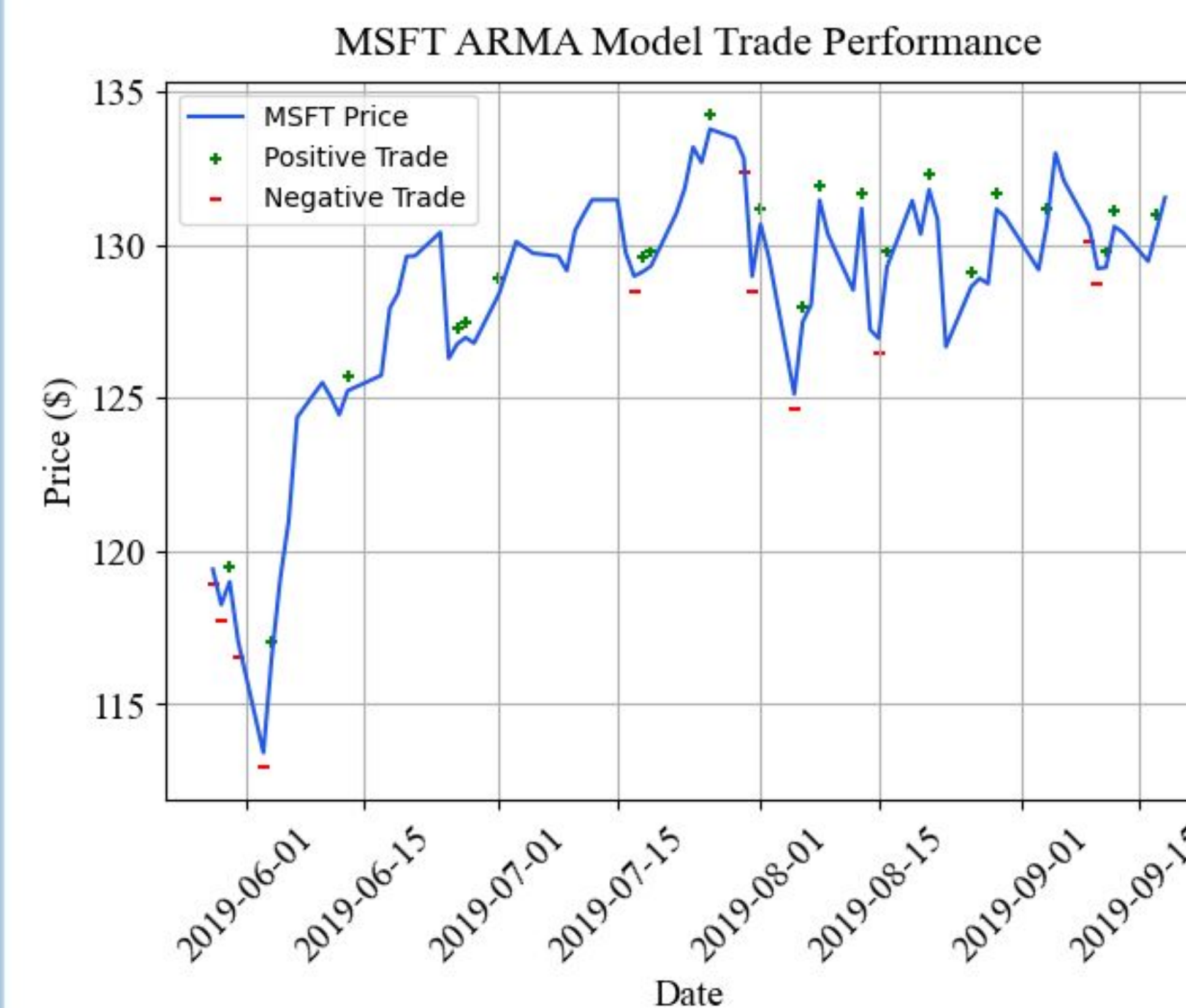
$$X_t = \sum_{i=1}^p \beta_i X_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \sum_{i=1}^P \phi_i X_{t-is} + \sum_{i=1}^Q \omega_i \epsilon_{t-is} + \epsilon_t$$

- The Random Forest (RF) model utilizes bootstrapping to generate multiple decision trees then aggregates their outputs to produce a prediction. The first 5 past values (lags) were used as features in producing the model's output.
- A single decision tree example is shown below:

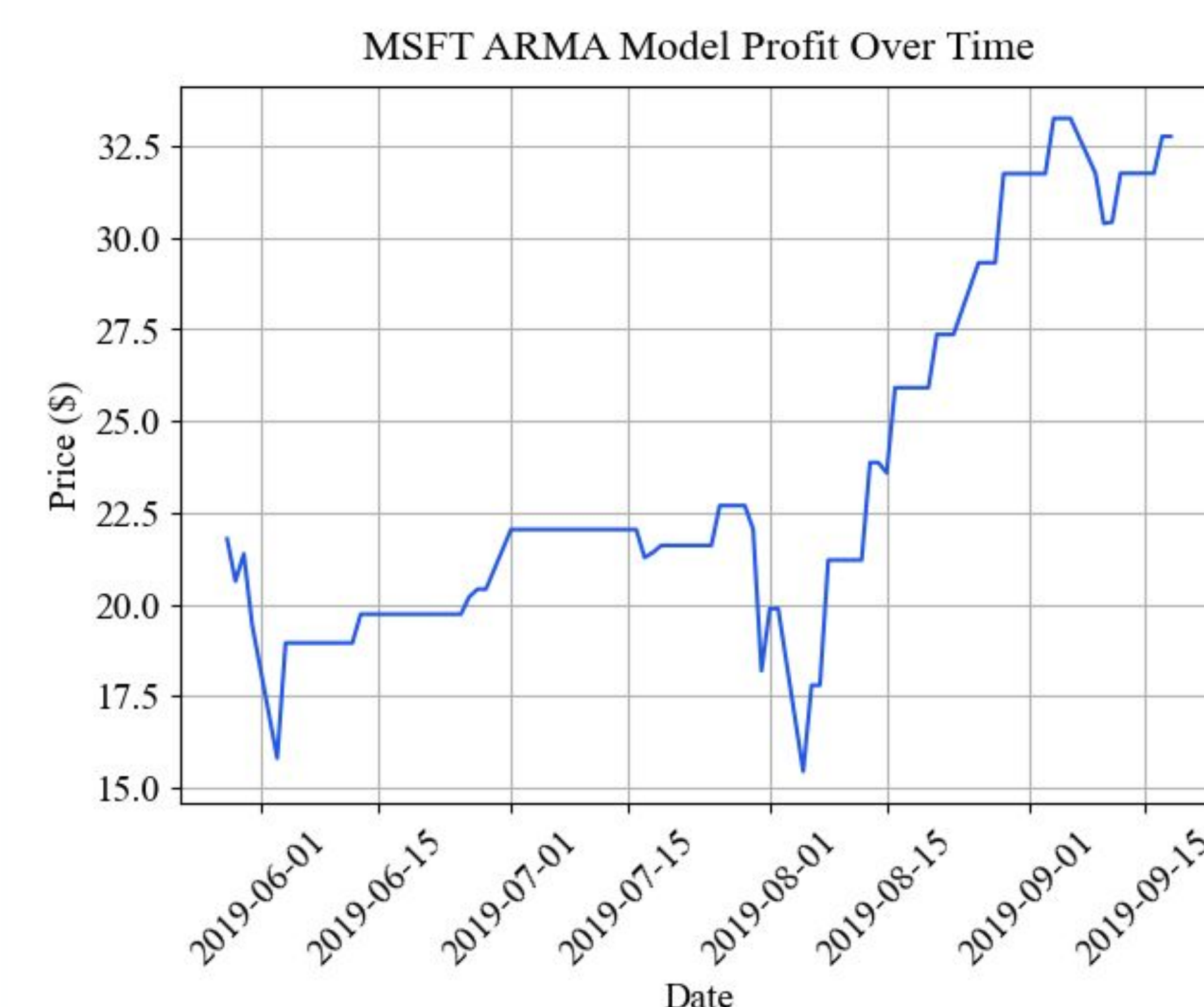


Simulation

The process to evaluate our models began with the models being trained on specific stock tickers closing prices from 2015-2017. The models' hyperparameters were then fine tuned using 2018 as a validation dataset to improve generalizability. Finally the models were evaluated using an inclusive rolling window on the 2019 test data with their profits over time being tracked. Trading on 2019 was simulated by making the models purchase when the predicted log return was above a certain threshold delta ($\delta = 0$ or 0.001) and sold at the next day's market closing price.



The plot above shows when the ARMA model conducted trades on MSFT from early June to mid September. The plot below shows the model's profit over time through the same time period.



Results

Model Name	MSFT Profit (2019)	TSLA Profit (2019)
AR ($\delta = 0$)	\$22.56	\$6.37
AR ($\delta = 0.001$)	\$13.98	-\$1.04
ARMA ($\delta = 0$)	\$35.42	-\$7.34
ARMA ($\delta = 0.001$)	\$32.07	-\$10.41
SARIMA ($\delta = 0$)	\$31.47	\$1.16
SARIMA ($\delta = 0.001$)	\$33.64	\$1.81
Random Forest ($\delta = 0$)	\$31.08	-\$4.47
Random Forest ($\delta = 0.001$)	\$31.26	-\$4.84
Momentum Trader	\$32.68	-\$1.75
Bernoulli Trader ($p = 0.5$)	\$27.40	\$2.73
Oracle Trader	\$170.63	\$51.32

- Momentum Trader: Used for benchmarking and comparison purposes, bought if the previous day's log return was positive.
- Bernoulli Trader: Used for benchmarking and comparison purposes, bought randomly with a probability of 0.5.
- Oracle Trader: Used for benchmarking and comparison purposes, maximum potential profits on each specific ticker.
- Tesla exhibited much more volatile behavior than Microsoft causing most of the models to perform more poorly on it.
- Microsoft exhibited a relatively stable log return behavior which allowed most of the time series models to perform better and more consistently relative to one another.
- Most of the models were extremely sensitive to δ which would vastly influence the model's trading performance.

Future Work

Future plans entail evaluating the models on different types of stock tickers and implementing exogenous variables that could potentially improve performance. Other avenues of exploration include investigating intra-day trading and evaluating the models on more complex/realistic simulated trading scenarios.

Acknowledgements

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